

Elevator System Architecture

Patterns: State Pattern, Strategy Pattern (SCAN/LOOK Algorithm), Observer Pattern | Domain: Resource Booking / Motion Control

Core Requirements & Features	Core Domain Classes & Interfaces
• Multi-Elevator Dispatch: Dispatch optimal elevator car based on floor requests. • Internal/External Calls: Handle passenger floor selection & hallway up/down calls. • Safety & State: Track Direction (UP, DOWN, IDLE) and Door Status (OPEN, CLOSED).	• ElevatorController: Manages car state, floor queues, and door operations. • ElevatorSystem: Central dispatcher allocating requests across multiple cars. • DispatchStrategy: Interface for FCFS, Shortest Seek Time, and LOOK algorithms. • ElevatorCar: Model for current floor, direction, weight limit, & doors.

Critical Execution Workflow & Method Trace

1. Passenger presses External Hall Call (floor=5, dir=UP).
 2. ElevatorSystem receives call and evaluates DispatchStrategy across all ElevatorCars.
 3. DispatchStrategy computes cost for each car: (Distance + Direction Penalty + Queue Length). Car moving towards request in same direction has lowest cost.
 4. Target ElevatorController adds floor 5 to its internal request queue (TreeSet / PriorityQueue).
 5. ElevatorCar moves floor by floor using LOOK algorithm: moves in current direction until no further requests exist in that direction, then reverses.
 6. Upon reaching floor 5, ElevatorCar changes State to STOPPED, opens doors, waits timer, closes doors, and resumes IDLE/MOVING.

Concurrency & Thread Safety Mechanics	Design Trade-offs & Interview FAQs
• Queue Safety: Thread-safe ConcurrentSkipListSet or PriorityBlockingQueue for up/down floor requests. • Car State Locks: Synchronized state transitions on ElevatorCar (IDLE -> MOVING_UP -> STOPPED). • Sensor Events: Asynchronous floor sensor callbacks executed on dedicated event threads.	• FCFS vs LOOK: FCFS causes high latency & starvation; LOOK (Elevator Algo) maximizes throughput. • Capacity Limits: Weight sensor triggers OVERLOADED state, bypassing external hall calls. • Decoupling: ElevatorController decoupled from ElevatorCar hardware drivers via interfaces.